

Research Summary

Operational Impact of Cross-Layer Telemetry Correlation in a Reproducible SOC Testbed

Overview

This research presents a controlled, systems-level evaluation of cross-layer telemetry correlation within a Security Operations Center (SOC) architecture. Rather than proposing a new intrusion detection algorithm, the study isolates correlation logic within the SIEM as the independent architectural variable while holding detection tooling constant.

The central objective is to quantitatively measure how correlating host-level telemetry (Sysmon and Wazuh agent logs) with network-level telemetry (Zeek logs) affects detection fidelity, alert noise, and detection latency compared to analyzing each source independently.

Modern SOC environments commonly ingest multiple telemetry streams but often evaluate them in partial isolation. While correlation is widely assumed to improve detection, its operational trade-offs—particularly in small-scale or resource-constrained environments—remain insufficiently characterized. This research investigates under what conditions correlation improves detection performance, when it provides neutral benefit, and when it may introduce latency or degradation.

Research Objectives

The study aims to:

- Compare detection performance across three configurations:
 - * Host-only detection
 - * Network-only detection
 - * Cross-layer correlated detection
- Quantify alert noise reduction using a formal Alert Consolidation Ratio (CR)
- Measure detection latency under each configuration
- Evaluate sensitivity to the correlation time window (Δt)

- Assess robustness under controlled operational stress (partial log loss)
- Demonstrate a reproducible experimental methodology for SOC architectural evaluation

Experimental Architecture

The evaluation is conducted within a reproducible four-node virtualized SOC testbed operating on an isolated internal network.

The environment consists of:

- Attacker System: Executes controlled intrusion scenarios
- Windows Endpoint: Instrumented with Sysmon and a Wazuh agent
- Zeek Sensor: Captures network telemetry and exports structured logs
- SIEM Platform: Wazuh stack (Manager, Indexer, Dashboard)

All configurations use identical logging pipelines and detection rules. The only variable modified across experiments is the presence or absence of cross-layer correlation within the SIEM. This ensures architectural isolation of the independent variable.

Correlation Definition

Correlation is formally defined as the association of host and network telemetry events that satisfy:

- Shared entity attributes (IP address, hostname, user, or process)
- Temporal proximity within a defined time window (Δt)
- Contextual linkage between process activity and network behavior

The correlation window (Δt) is evaluated at 10 seconds, 30 seconds, and 60 seconds to analyze trade-offs between alert consolidation and detection latency.

This study evaluates correlation as a configuration choice within the SIEM, not as a newly designed detection engine.

Attack Scenarios

Experiments include structured and repeatable intrusion scenarios:

- **Reconnaissance and authentication brute-force activity**
- **Multi-stage attack chain involving privilege escalation and lateral movement**
- **Living-off-the-land (LOTL) activity using native system tools (e.g., PowerShell abuse, scheduled tasks)**

Each scenario is executed multiple times under identical conditions to enable comparative measurement.

Ground truth timestamps are recorded at the attacker system to formally validate detection timing and classification.

Controlled Operational Stress

To evaluate robustness, the study introduces one controlled failure condition:

- Partial network log loss

This stress test evaluates how degraded telemetry impacts detection accuracy, alert consolidation, and latency.

The goal is to characterize correlation behavior under constrained operational conditions, not to simulate enterprise-scale infrastructure.

Evaluation Metrics

Detection performance is evaluated using:

- True Positive Rate (TPR)
- False Positive Rate (FPR)
- Detection Latency

- Alert Consolidation Ratio (CR)

The Alert Consolidation Ratio is formally defined as:

$$CR = 1 - (N_c / (N_h + N_n))$$

Where:

- N_h = Host-only alerts
- N_n = Network-only alerts
- N_c = Correlated detection events

This metric quantifies the proportional reduction of independent alert noise into consolidated correlated incidents.

True positives and false positives are defined relative to recorded ground truth attack timestamps within the selected Δt window.

Threats to Validity

The study acknowledges limitations related to:

- Small-scale four-node architecture
- Reduced log volume relative to enterprise SOC environments
- Simplified user behavior
- Controlled experimental conditions

The objective is architectural trade-off evaluation under controlled conditions, not enterprise benchmarking.

Expected Contribution

This research contributes:

- A reproducible framework for evaluating SOC correlation architectures
- A formalized methodology for measuring operational trade-offs
- Empirical characterization of how correlation window selection impacts detection

fidelity and alert noise

- Insight into correlation robustness under telemetry degradation

The study emphasizes experimental rigor, variable isolation, and reproducibility rather than algorithmic novelty. It demonstrates structured systems thinking and quantitative evaluation of SOC architecture decisions.